ELECTRONIC SUPPLEMENTARY INFORMATION (ESI)

A Spacetime Algebra Framework for Passive Quench Protection and Radial Shunting in Non-Insulated REBCO High-Field Fusion Magnets

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S1. Mathematical Derivations & First-Principles Proofs

In $Cl_{1,3}$ Spacetime Algebra, the multivector rotor $R = \exp(-\frac{1}{2}\,\Omega\,\tau)$ preserves localized energy positivity $T_{00} = \frac{1}{2}\,(\varepsilon_0 E^2 + \mu_0^{-1} B^2) \geq 0$. Current continuity $\nabla \cdot \mathbf{J} = 0$ across turn-to-turn contact surface resistivity $\rho_{ct} = 15.0\ \mu\Omega\cdot cm^2$ governs radial current bypass fractions during localized quenches.

S2. Thermophysical & Material Property Matrix

Parameter / Metric	Design Value	Safety Status / Margin
REBCO Critical Current J_c (4.2 K, 20 T)	> 10^3 A/mm²	High Field Benchmark
Turn-to-Turn Contact Resistivity ρ_ct	15.0 μΩ·cm²	Calibrated Radial Shunt
L/R Circuit Time Constant τ_L/R	1.0 – 3.0 s	Passive Current Decay
Max Adiabatic Quench Temp T_max	185 K ≤ 200 K	Unconditional Self-Protection
Reversible Strain Window e_ax	-0.30% to +0.35%	Structural Elastic Margin